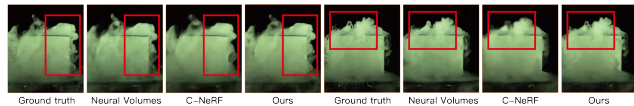


Representing Volumetric Videos as Dynamic MLP Maps

We thank the reviewers for their insightful feedback and finding our method novel (R1), interesting (R2), practically valuable (R3), and results promising (R1, R2, R3). We address all the comments below and will revise the paper.

Common Questions

CQ1: Results on data of NV. We performed experiments on a video of NV, showing that our method outperforms NV and C-NeRF (LPIPS: 0.198 vs. 0.218 and 0.227). Below are some visual results. We will include the comparison.



CQ2: Ablation studies of hash tables. We conducted two ablation studies on “Twirl” of ZJU-MoCap, which replace hash tables with sinusoid encoding [37] and tri-plane feature maps [5], respectively. Results show that hash tables perform better (LPIPS: 0.049 vs. 0.068 and 0.055).

Responses to Reviewer 1 (EayZ)

Q1: Projection of 3D points. We project 3D points into MLP maps for querying MLP parameters (lines 309-318).

Q2: Failure cases. Fig. F1 (a) presents some failure cases, where we struggle to render the text and fingers.

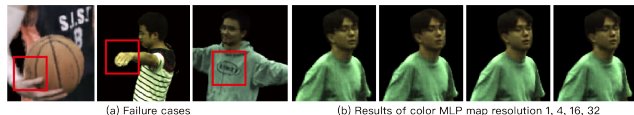


Figure F1. Results on the ZJU-MoCap and NHR datasets.

Responses to Reviewer 2 (6Djx)

Q1: Missing details. We presented some details on rendering using the MLP maps (lines 309-320 and Sec.1 of Supp.) and loss function (lines 480-481). The MSE loss is used for training in practice. We will include the complete rendering pipeline and detailed loss function in the revised paper and release the code for reproducibility.

Q2: Latent space. We agree that ‘latent space’ is an overused term and will change it.

Q3: Reasons of higher quality than baselines. Tab. T1 indicates two reasons: 1) Hash tables (Ours-Single vs. C-NeRF). 2) MLP maps (Ours vs. Ours-Single), which allow each MLP to represent a small region, in contrast to the single NeRF modeling the whole volumetric video.

Methods	Baseline1	C-NeRF	Ours-Single	Ours
Description	feature tri-planes + single NeRF	feature volumes + single NeRF	hash tables + single NeRF	hash tables + MLP maps
LPIPS	0.065	0.057	0.056	0.049

Table T1. Comparison on the “Twirl” subject of ZJU-MoCap.

Q4: Why increasing color map resolution does not always increase visual quality. Since this will make the training more difficult for two reasons. First, it will increase the number of parameters. Second, increasing the resolution need to decrease the number of query points fed into

each color MLP during each training iteration due to memory limit, thereby decreasing the batch size. Fig. F1 (b) shows the visual results of different color map resolutions.

Q5: Storage cost. We will include this limitation.

Q6: Related work on tri-planes. We cited these related methods in line 207. The difference is that we use tri-plane MLP maps while they use tri-plane feature maps, and we can model the 3D scene with a single MLP map instead of three. We will expand the discussion in the revised paper.

Q7: Different kinds of content. Please refer to CQ1.

Q8: Rendering speed of Neural Volumes. We used the official code of Neural Volumes (NV) for comparison. NV takes 49 ms for rendering an image on ZJU-MoCap, which is close to real-time rendering.

Q9: Ablation studies of hash tables. Please refer to CQ2.

Responses to Reviewer 3 (Le24)

Q1: Motivation of using two parameter sets. We use a hybrid of hash tables and MLP maps to achieve both high quality and efficiency. 1) Why MLP maps: Only using hash tables needs a relatively large MLP to achieve high quality, which will be slow due to the costly network evaluation, as suggested by the rendering speed of DyNeRF and C-NeRF. Using MLP maps increases the rendering speed with small MLPs. 2) Why hash tables: compared to MLP maps alone, hash tables improve the rendering quality, as shown in CQ2. We will make the motivation clearer in the revised paper.

Q2: Why use MLP maps instead of feature maps. As stated in Q1, we use MLP maps to improve rendering speed. Tab. T1 indicates that using feature maps instead of MLP maps needs a large MLP to achieve high quality, which will be slow. Feature maps can work together with MLP maps, but it does not perform better than hash tables (CQ2).

Q3: Baseline methods not optimized on tested data. The focus of this work is not the rendering quality, but the rendering and storage efficiency which should be similar across datasets. In terms of quality, we provide additional results on the data of NV [33] as shown in CQ1.

Q4: Contribution to rendering speed. ESS is a super effective technique. We believe that achieving another 4x speedup on top of it by a novel representation design is non-straightforward, beneficial to the community, and critical for real-time applications. In addition, the rows 6 and 9 in Tab. 1 shows that MLP maps make the rendering 9x faster when the ESS is not used.

Q5: High-resolution density MLP maps. As stated in lines 461-469, density MLP has 32 parameters, and color MLP has 2624 parameters, which means that 256×256 density MLP map has similar number of parameters to 28×28 color MLP map. In addition, the scene geometry generally has lower frequency than the appearance. Therefore, the high-resolution density MLP maps are easier to train.